

JRCAI

مركز الأبحاث المشترك للذكاء الاصطناعي

Joint Research Center for AI

SDAIA | KFUPM

King Fahd University of Petroleum and Minerals (KFUPM)

ASSIGNMENT COVER SHEET

Student Name	Student ID
Ahmed Almutairi	

Assignment Details

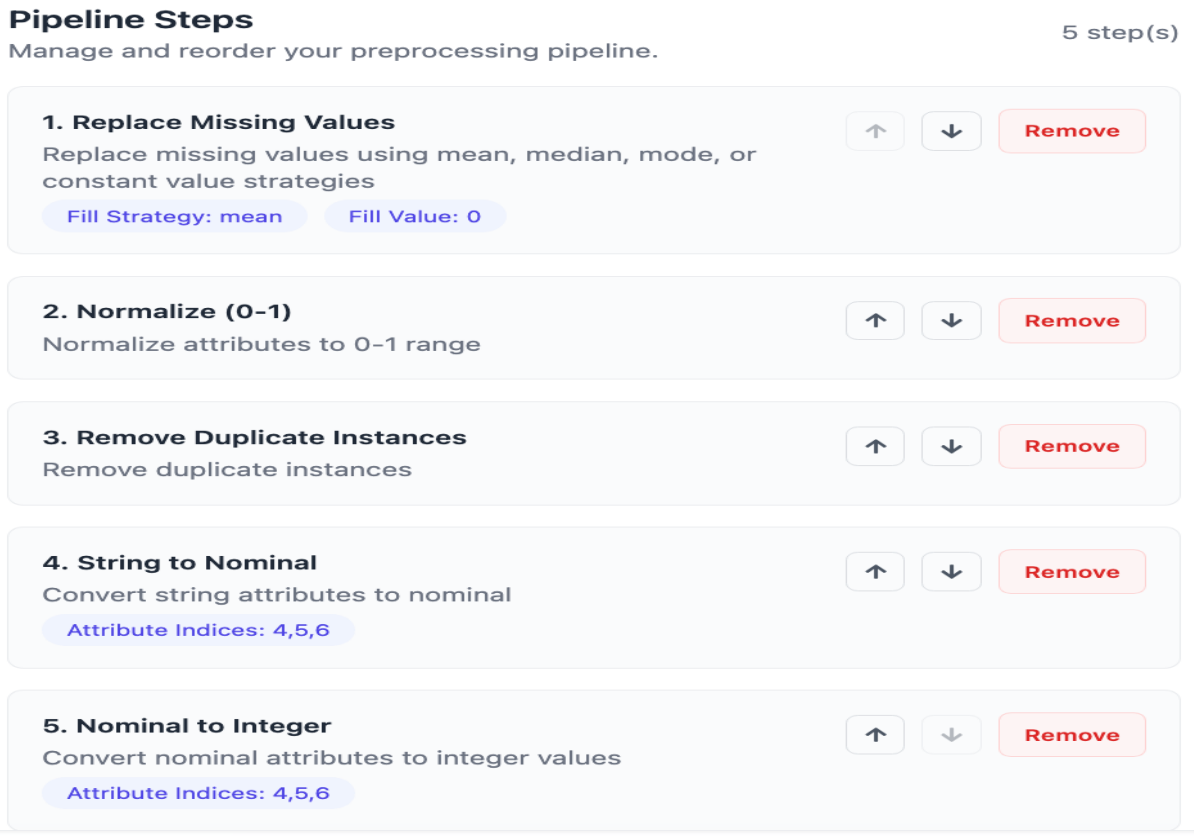
To be marked by: Dr. Safwan Mahmood Al-Selwi	
Date submitted:	Submission Due Date: 24/7/2026
Submission of this assignment agrees to the following: *I understand that the piece of work submitted will be considered as the final and complete version of my assignment of which I am the sole author. I understand both the meaning and consequences of plagiarism and that my work has been appropriately attributed unless otherwise stated. I have not knowingly allowed another to copy my work.*	

Assignment Feedback - Lecturer Use Only

First Marker Feedback:	
Signature of first marker:	**% Mark** *(Un-moderated)*

Phase 2 Task 1 - Data Preparation

- Using Weka AI – (This Pipeline have been used)



The Results is as Shown:

INCOME	GENDER	PRODUCTCATEGORY	PURCHASED
0	0	0	0
0.42857142857142855	1	1	1
0.4901098901098902	1	0	0
0.8571428571428571	0	2	0
0.14285714285714285	1	0	1

Using Python – The code that have been used:

```
import pandas as pd

# Load dataset
df = pd.read_csv('CustomerPurchaseDataset.csv')

# 1. Replace Missing Values (mean strategy for numeric columns)
df.fillna(df.mean(numeric_only=True), inplace=True)

# 2. Normalize attributes (0-1 range) for numerical columns
num_cols = ['Age', 'Income']
df[num_cols] = (df[num_cols] - df[num_cols].min()) / (df[num_cols].max() -
df[num_cols].min())

# 3. Remove Duplicate Instances
df.drop_duplicates(inplace=True)

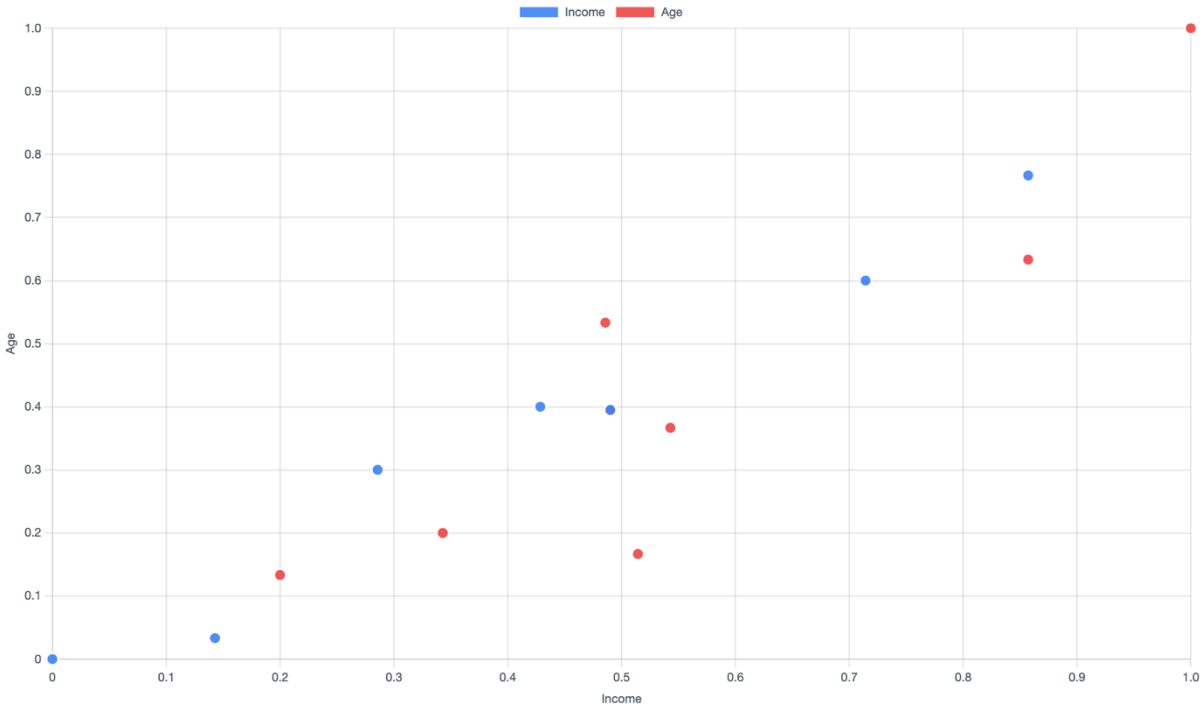
# 4 & 5. String to Nominal -> Nominal to Integer (Attributes 4, 5, 6: Gender,
ProductCategory, Purchased)
cat_cols = ['Gender', 'ProductCategory', 'Purchased']
for col in cat_cols:
    df[col] = df[col].astype('category').cat.codes

# Save preprocessed dataset
df.to_csv('preprocessed_CustomerPurchaseDataset.csv', index=False)
print(df)
```

Results is as Shown:

	CustomerID	Age	Income	Gender	ProductCategory	Purchased
0	1	0.000000	0.000000	1	2	1
1	2	0.400000	0.428571	0	1	0
2	3	0.394872	0.490110	0	2	1
3	4	0.766667	0.857143	1	3	1
4	5	0.033333	0.142857	0	2	0
5	6	0.600000	0.714286	1	0	1
6	7	0.300000	0.285714	0	3	0
7	8	0.166667	0.514286	1	1	1
8	9	1.000000	1.000000	0	0	1
9	10	0.533333	0.485714	1	1	0
10	11	0.133333	0.200000	0	3	1
11	12	0.394872	0.490110	1	2	1
12	13	0.200000	0.342857	1	0	0
13	14	0.633333	0.857143	0	3	1
14	15	0.366667	0.542857	1	2	0

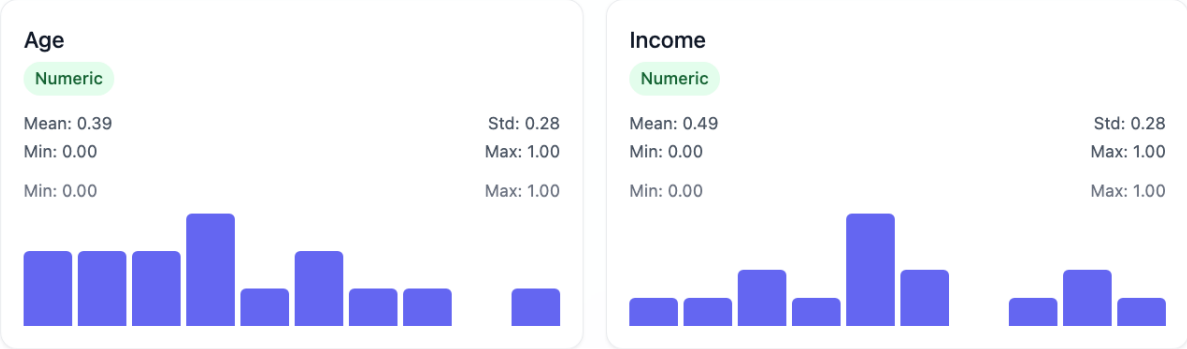
Visualization



The technique I have used is classification as in the end we need weather the customer will buy or not. The pre-processing steps are:

1. Repleace missing values with mean
2. Normalize the numbers (0-1)
3. If there is duplicates remove it
4. String to Numerical

The mean age is 33.8.
The mean income is 47153.8.



Model had low accuracy due to it having only 15 instances, if there was enough instance that the model (Decision tree classifier) can train on it would be very helpful.